

§ 3.6 中心力场中粒子运动的一般性质

一、中心力场的特点

- 势函数球对称的力场

中心力场的势能函数 $V(r)$ 与方向 θ, φ 无关, 具有转动对称性.

$$\hat{H} = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r), \quad r = |\vec{r}|$$

- 中心力场体系的轨道角动量守恒

$$\left[\hat{l}, \hat{H} \right] = 0$$

【思考1】 如何求解粒子在中心势场中的运动？

二、力学量完全集

力学量完全集一般选 $\{\hat{H}, \hat{L}^2, \hat{L}_z\}$, 用 ψ_{nlm} 代表共同本征态, 本征值问题表示为

$$\hat{H}\psi_{nlm} = E_{nl}\psi_{nlm}$$

$$\hat{L}^2\psi_{nlm} = l(l+1)\hbar^2\psi_{nlm}$$

$$\hat{L}_z\psi_{nlm} = m\hbar\psi_{nlm}$$

三、径向方程及其求解

1. 中心力场中运动粒子的Hamilton量

$$\hat{H} = \frac{p^2}{2\mu} + V(r) = -\frac{\hbar^2}{2\mu} \nabla^2 + V(r)$$

在球坐标系中

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} r^2 \frac{\partial}{\partial r} + \frac{1}{r^2} \nabla_{\theta\varphi}^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \nabla_{\theta\varphi}^2$$

$$\frac{p_r^2}{2\mu} = -\frac{\hbar^2}{2\mu} \left[\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right] = -\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{\partial^2}{\partial r^2} r$$

— 径向动能算符

$$\nabla_{\theta\varphi}^2 = \frac{1}{\sin\theta} \frac{\partial}{\partial\theta} (\sin\theta \frac{\partial}{\partial\theta}) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\varphi^2} = -\frac{\hat{L}^2}{\hbar^2}$$

$$\hat{H} = \frac{p_r^2}{2\mu} + \frac{\hat{L}^2}{2\mu r^2} + V(r)$$

2. 能量本征方程及其求解

$$\left[-\frac{\hbar^2}{2\mu} \frac{1}{r} \frac{\partial^2}{\partial r^2} r + \frac{\hat{L}^2}{2\mu r^2} + V(r)\right] \Psi(r, \theta, \phi) = E \Psi(r, \theta, \phi)$$

因势函数球对称则可分离变量

$$\Psi(r, \theta, \phi) = R(r) Y_{lm}(\theta, \phi), \quad l = 0, 1, 2, \dots$$

$$m = l, l-1, \dots, -l$$

代入能量本征方程, 得径向方程

$$\left[\frac{1}{r} \frac{d^2}{dr^2} r + \frac{2\mu}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2}\right] R_l(r) = 0$$

令 $\chi_l(r) = r R_l(r)$, 径向方程化为

$$\chi_l''(r) + \left[\frac{2\mu}{\hbar^2} (E - V(r)) - \frac{l(l+1)}{r^2}\right] \chi_l(r) = 0$$

【讨论】

- 中心力场本征值问题的主要计算任务，是针对给定的势函数求解上述径向方程.
- 中心力场能级的简并度一般为 $2l + 1$
- 中心势的定义域 $r \geq 0$

一维势 $V(x)$ 的定义域 $-\infty < x < +\infty$

3. 解的渐近行为

$$\lim_{r \rightarrow 0} R_l(r) \sim r^l$$

或 $\lim_{r \rightarrow 0} \chi_l(r) \sim r^{l+1}, \quad \chi_l(0) = 0$

【证明】

粒子出现在半径为 $r \rightarrow 0$ 的球体内的概率

$$\int_0^r R_l^2(r) r^2 dr \sim r^3 R_l^2(r)$$

统计诠释要求 $\lim_{r \rightarrow 0} r^3 R_l^2(r) \rightarrow 0$, 因此

$$\lim_{r \rightarrow 0} R_l(r) \sim r^S, \quad S > -\frac{3}{2}$$

当 $r \rightarrow 0$ 时, 一般 $V(r)$ 比 $\frac{1}{r^2} \rightarrow \infty$ 慢, 即 $\lim_{r \rightarrow 0} r^2 V(r) = 0$, 则径向方程的渐近形式为

(注: 通常碰到的中心场均满足此条件. 譬如, 谐振子势、线性中心势、对数中心势、球方势、自由粒子、库仑势、汤川势等)

$$\left[\frac{1}{r} \frac{d^2}{dr^2} r - \frac{l(l+1)}{r^2} \right] R_l(r) = 0$$

在附近 $r = 0$ 设 $R_l(r) \sim r^\lambda$, 代入渐近形式得

$$\lambda(\lambda+1) - l(l+1) = 0 \qquad R_l(r) \sim \begin{cases} r^l \\ r^{-(l+1)} \end{cases}$$

下面证明, 只有 $R_l(r) \sim r^l$ 才是正确的渐近解.

- 当 $l \geq 1$ 时, 只有 $R_l(r) \sim r^l$ 满足条件

$$\lim_{r \rightarrow 0} R_l(r) \sim r^S, \quad S > -\frac{3}{2}$$

- 当 $l=0$ 时, 虽然 $R_l(r) \sim r^{-(l+1)} = r^{-1}$ 也满足条件, 但这时 $\Psi \sim R_0(r)Y_{00} \sim \frac{1}{r}$, 因 $\nabla^2 \frac{1}{r} = -4\pi\delta(\vec{r})$, 则不满足Schrödinger方程

$$\left[-\frac{\hbar^2}{2\mu}\nabla^2 + V(r)\right]\Psi = E\Psi$$

【注: $\nabla^2 \frac{1}{r} = -4\pi\delta(\vec{r})$ 的证明见本节的附录】

4. 离心势能

r 方向的有效势为
$$V_{\text{eff}} = V(r) + \frac{\hat{L}^2}{2\mu r^2} \Rightarrow V_{\text{eff}} = V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2}$$

离心势能(centrifugal potential)

$$U_l(r) = \frac{l(l+1)\hbar^2}{2\mu r^2} \geq 0 \quad U_0(r) = 0$$

【理由】 它在 $r=0$ 附近构筑了很高的势垒, 产生自中心向外的斥力, 使粒子在 $r=0$ 附近出现的概率明显下降(后面知, 当 $r \rightarrow 0$ 时, $R(r)$ 将以 $r^l \rightarrow 0$), 而且 l 越大这种现象越明显.

角动量量子数用下面记号代表

$$l = 0, \quad 1, \quad 2, \quad \dots$$

$$s, \quad p, \quad d, \quad \dots$$

5. 宇称

在空间反射下, 矢量 $\vec{r} \rightarrow -\vec{r}$, 在球坐标系中

$$\vec{r}(r, \theta, \varphi) \Rightarrow -\vec{r}(r, \pi - \theta, \pi + \varphi)$$

$$Y_{lm}(\theta, \varphi) = N_{lm} P_l^m(\cos \theta) \exp(im\varphi)$$

$$P_l^m[\cos(\pi - \theta)] = P_l^m(-\cos \theta) = (-1)^{l-m} P_l^m(\cos \theta)$$

$$e^{im(\pi + \varphi)} = (-1)^m e^{im\varphi}$$

$$Y_{lm}(\pi - \theta, \pi + \varphi) = (-1)^l Y_{lm}(\theta, \varphi)$$

宇称由 $Y_{lm}(\theta, \varphi)$ 决定, 因此本征态的宇称为 $(-1)^l$

【思考2】 如何将解决的实际问题转化为粒子在中心势场中的运动问题？

四、两体问题的求解

质量为 m_1 和 m_2 两粒子体系的Schrödinger方程

$$i\hbar \frac{\partial}{\partial t} \Psi(x_1, y_1, z_1; x_2, y_2, z_2; t) \\ = \left[-\frac{\hbar^2}{2m_1} \left(\frac{\partial^2}{\partial x_1^2} + \frac{\partial^2}{\partial y_1^2} + \frac{\partial^2}{\partial z_1^2} \right) - \frac{\hbar^2}{2m_2} \left(\frac{\partial^2}{\partial x_2^2} + \frac{\partial^2}{\partial y_2^2} + \frac{\partial^2}{\partial z_2^2} \right) + V(|\vec{r}_1 - \vec{r}_2|) \right] \Psi(x_1, y_1, z_1; x_2, y_2, z_2; t)$$

引入相对坐标

$$\vec{r} = \vec{r}_1 - \vec{r}_2$$

$$x = x_1 - x_2, \quad y = y_1 - y_2, \quad z = z_1 - z_2$$

质心坐标

$$\vec{R} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{M}$$

$$MX = m_1 x_1 + m_2 x_2, \quad MY = m_1 y_1 + m_2 y_2, \quad MZ = m_1 z_1 + m_2 z_2$$

质心质量

$$M = m_1 + m_2$$

约化质量

$$\mu \equiv \frac{m_1 m_2}{m_1 + m_2}$$

$$\frac{\partial}{\partial x_1} = \frac{\partial X}{\partial x_1} \frac{\partial}{\partial X} + \frac{\partial x}{\partial x_1} \frac{\partial}{\partial x} = \frac{m_1}{M} \frac{\partial}{\partial X} + \frac{\partial}{\partial x}$$

$$\begin{aligned} \frac{\partial^2}{\partial x_1^2} &= \left(\frac{m_1}{M} \frac{\partial}{\partial X} + \frac{\partial}{\partial x} \right) \left(\frac{m_1}{M} \frac{\partial}{\partial X} + \frac{\partial}{\partial x} \right) \\ &= \frac{m_1^2}{M^2} \frac{\partial^2}{\partial X^2} + \frac{2m_1}{M} \frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2} \end{aligned}$$

$$\frac{\partial}{\partial x_2} = \frac{\partial X}{\partial x_2} \frac{\partial}{\partial X} + \frac{\partial x}{\partial x_2} \frac{\partial}{\partial x} = \frac{m_2}{M} \frac{\partial}{\partial X} - \frac{\partial}{\partial x}$$

$$\begin{aligned} \frac{\partial^2}{\partial x_2^2} &= \left(\frac{m_2}{M} \frac{\partial}{\partial X} - \frac{\partial}{\partial x} \right) \left(\frac{m_2}{M} \frac{\partial}{\partial X} - \frac{\partial}{\partial x} \right) \\ &= \frac{m_2^2}{M^2} \frac{\partial^2}{\partial X^2} - \frac{2m_2}{M} \frac{\partial^2}{\partial X \partial x} + \frac{\partial^2}{\partial x^2} \end{aligned}$$

可以证明

$$\frac{1}{m_1} \nabla_1^2 + \frac{1}{m_2} \nabla_2^2 = \frac{1}{M} \nabla_R^2 + \frac{1}{\mu} \nabla^2$$

$$\nabla_R^2 = \frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2}$$

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

Schrödinger方程变成

$$i\hbar \frac{\partial}{\partial t} \Psi(X, Y, Z; x, y, z; t)$$

$$= - \left[\frac{\hbar^2}{2M} \left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2} \right) + \frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) + V(\vec{r}) \right] \Psi(X, Y, Z; x, y, z; t)$$

设 $\Psi = \psi(x, y, z) \phi(X, Y, Z) \chi(t)$

$$\left\{ \begin{array}{l} \frac{i\hbar}{\chi} \frac{d\chi}{dt} = E_t \\ -\frac{\hbar^2}{2\mu} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \right) \psi + V(x, y, z) \psi = E \psi \\ -\frac{\hbar^2}{2M} \left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2} \right) \phi = E_C \phi \end{array} \right. \quad (E_C = E_t - E)$$

1. 含时间部分的解

$$\chi(t) = Ce^{-\frac{i}{\hbar}E_t t}$$

2. 单体问题的解

$$-\frac{\hbar^2}{2M}\left(\frac{\partial^2}{\partial X^2} + \frac{\partial^2}{\partial Y^2} + \frac{\partial^2}{\partial Z^2}\right)\phi = (E_t - E)\phi$$

-质心运动状态的波函数所满足的方程

质心以质量M、能量为 $E_C (= E_t - E)$ 的自由粒子的方式运动, 即平面波, 与内部性质无关.

E_C : 质心运动能量

3. 相对运动

$$-\frac{\hbar^2}{2\mu}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)\psi + V(x, y, z)\psi = E\psi$$

-相对运动方程

E : 相对运动能量

【附录】证明 $\nabla^2 \frac{1}{r} = -4\pi\delta(\vec{r})$

$$\begin{aligned} \exp(i\vec{k} \cdot \vec{r}) &= -\frac{\nabla^2 \exp(i\vec{k} \cdot \vec{r})}{k^2} & \delta(\vec{r}) &= \frac{1}{(2\pi)^3} \int_{-\infty}^{\infty} \exp(i\vec{k} \cdot \vec{r}) d^3k \\ \int_{-\infty}^{\infty} \frac{\exp(i\vec{k} \cdot \vec{r})}{k^2} d^3k & & &= \frac{-1}{(2\pi)^3} \int_{-\infty}^{\infty} \frac{\nabla^2 \exp(i\vec{k} \cdot \vec{r})}{k^2} d^3k \\ &= \int_0^{2\pi} d\phi \int_0^{\pi} \sin\theta d\theta \int_0^{\infty} \frac{\exp(ikr \cos\theta)}{k^2} k^2 dk & &= \frac{-1}{(2\pi)^3} \nabla^2 \int_{-\infty}^{\infty} \frac{\exp(i\vec{k} \cdot \vec{r})}{k^2} d^3k \\ &= 4\pi \int_0^{\infty} dk \frac{\sin kr}{kr} = 2\pi^2 \frac{1}{r} \end{aligned}$$

$$\begin{aligned} \delta(\vec{r}) &= \frac{-1}{(2\pi)^3} \nabla^2 \int_{-\infty}^{\infty} \frac{\exp(i\vec{k} \cdot \vec{r})}{k^2} d^3k \\ &= \frac{-1}{(2\pi)^3} 2\pi^2 \nabla^2 \frac{1}{r} \end{aligned}$$

